

Diksha Singh

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SUMMARY

Data Scientist and independent researcher specialising in ML reliability, data quality engineering, and algorithmic trading systems. Founder of NXIO, an algorithmic trading intelligence platform. Built production-grade ML pipelines — label noise detection, prediction calibration, and feature stability diagnostics — using Python, FastAPI, PostgreSQL, and Streamlit.

TECHNICAL SKILLS

Languages: Python, SQL, Bash

ML / Data Science: Scikit-learn, Pandas, NumPy, SciPy, XGBoost, SHAP, Label Noise Detection, Model Calibration, Feature Importance Analysis, Statistical Hypothesis Testing, A/B Testing, Time-Series Analysis

Backend / Infra: FastAPI, REST APIs, PostgreSQL, SQLite, Streamlit, Docker (basics), Git, GitHub

Visualisation: Matplotlib, Seaborn, Plotly, Microsoft Power BI

Domain: Algorithmic Trading, NSE/BSE/MCX, SEBI Compliance, Quantitative Finance, ML Observability, Data Quality Engineering

EXPERIENCE

Founder & Lead Data Scientist Jul 2025 – Present
NXIO — Algorithmic Trading Intelligence Platform *Delhi, India*

- Architected an end-to-end market intelligence system covering NSE, BSE, and MCX with real-time data pipelines, algorithmic signal generation, and backtesting modules.
- Built quantitative trading strategies using statistical models for price prediction and risk assessment; ensured compliance with SEBI's April 2026 Algo-ID mandate.
- Designed automated feature engineering pipelines handling multi-exchange tick data with anomaly detection and alerting.

Independent ML Researcher Jun 2024 – Present
Self-Directed Research — ML Reliability & Data Quality *Remote*

- Prototyped research projects targeting label noise, model calibration, and feature stability — underserved areas in current ML tooling.
- Authored formal research documentation, technical papers, and reproducible experiment designs with ablation studies.

PROJECTS

Label Quality & Noise Engine (LQNE) | *Python, Scikit-learn, Pandas, Git* Dec 2025 – Feb 2026
– Built pipeline to detect mislabelled samples using cross-validation confidence thresholding and ensemble disagreement scoring; published on GitHub as a reusable data quality module.

Prediction Confidence Decay Tracker | *FastAPI, Streamlit, PostgreSQL, SciPy* 2025
– Engineered post-deployment calibration monitoring system tracking confidence degradation over time with automated drift alerts; Streamlit dashboard visualising ECE and reliability curves.

Feature Importance Instability Detector | *Python, Scikit-learn, SHAP, NumPy* 2025
– Designed diagnostic tool flagging instability in SHAP/permutation feature rankings across bootstrap samples for production ML reliability.

HACKATHONS & CERTIFICATIONS

E-Raksha Hackathon 2026 | *IIT Delhi — National Cybersecurity Competition* 2026
Bharatiya Antariksh Hackathon 2025 | *ISRO, Powered by Hack2Skill — Acknowledged for Innovation* 2025
redBus Data Decode Hackathon 2025 | *Demand Forecasting, RMSE-evaluated leaderboard* 2025
Basics of Microsoft Power BI | *UniAthena × Cambridge International Qualifications (CIQ) — Blockchain Verified* Jan 2026

EDUCATION

Chaudhary Charan Singh (CCS) University Sep 2024 – Apr 2027
Bachelor of Computer Applications (BCA) — Computer and Information Sciences *Meerut, India*
– Coursework: Data Structures, Algorithms, DBMS, Computer Networks, Mathematics III